



Ordering negative integers

Task A: Ordering negative integers in a context

1) Write the following temperatures from coldest to warmest.

- a) $(-10)^{\circ}\text{C}$ $(-4)^{\circ}\text{C}$ $(-8)^{\circ}\text{C}$ $(-2)^{\circ}\text{C}$ $(-5)^{\circ}\text{C}$
 $(-10)^{\circ}\text{C}$ $(-8)^{\circ}\text{C}$ $(-5)^{\circ}\text{C}$ $(-4)^{\circ}\text{C}$ $(-2)^{\circ}\text{C}$
- b) $(-18)^{\circ}\text{C}$ $(-22)^{\circ}\text{C}$ $(-24)^{\circ}\text{C}$ $(-16)^{\circ}\text{C}$ $(-27)^{\circ}\text{C}$
 $(-27)^{\circ}\text{C}$ $(-24)^{\circ}\text{C}$ $(-22)^{\circ}\text{C}$ $(-18)^{\circ}\text{C}$ $(-16)^{\circ}\text{C}$
- c) $(-32)^{\circ}\text{C}$ $(-23)^{\circ}\text{C}$ $(-43)^{\circ}\text{C}$ $(-34)^{\circ}\text{C}$ $(-48)^{\circ}\text{C}$
 $(-48)^{\circ}\text{C}$ $(-43)^{\circ}\text{C}$ $(-34)^{\circ}\text{C}$ $(-32)^{\circ}\text{C}$ $(-23)^{\circ}\text{C}$
- d) $(-100)^{\circ}\text{C}$ $(-104)^{\circ}\text{C}$ $(-96)^{\circ}\text{C}$ $(-88)^{\circ}\text{C}$ $(-110)^{\circ}\text{C}$
 $(-110)^{\circ}\text{C}$ $(-104)^{\circ}\text{C}$ $(-100)^{\circ}\text{C}$ $(-96)^{\circ}\text{C}$ $(-88)^{\circ}\text{C}$

2) A quiz has ten questions.

Points are given as follows: 2 points for each correct answer

(-1) for a missed question

(-3) for an incorrect answer

I scored (-9)



Lucas

I scored (-3)



Laura

I scored (-10)



Jacob

I scored (-5)



Sofia

Work out their score and rank them from lowest to highest score.

Jacob (-10) , Lucas (-9) , Sofia (-5) and Laura (-3)

3) A hotel has 10 floors below reception which is on the ground floor, floor 0.

There are bedrooms only on floors (-2) , (-4) , (-6) and (-7) .

On the other floors are the following: staff room, gym, restaurant, maintenance office, basement and swimming pool.

Use the following clues to work out what is on each floor.

- a) The swimming pool is 2 floors below the gym.
 b) The staff room is between the basement and maintenance office.
 c) The restaurant is higher up than the gym.
 d) The maintenance office is 7 floors below the restaurant.

FLOOR	
0	Reception
(-1)	Restaurant
(-2)	Bedrooms
(-3)	Gym
(-4)	Bedrooms
(-5)	Pool
(-6)	Bedrooms
(-7)	Bedrooms
(-8)	Office
(-9)	Staff room
(-10)	Basement



Task B: Ordering negative integers

1) Write the following sets of integers in ascending order.

a) $(-6), (-9), (-5), (-8), (-3)$
 $(-9), (-8), (-6), (-5), (-3)$

b) $(-18), (-19), (-25), (-28), (-13)$
 $(-28), (-25), (-19), (-18), (-13)$

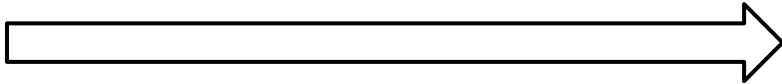
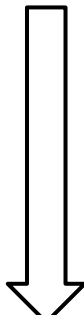
c) $(-89), (-91), (-88), (-99), (-19)$
 $(-99), (-91), (-89), (-88), (-19)$

d) $(-256), (-119), (-112), (-265), (-312)$
 $(-312), (-265), (-256), (-119), (-112)$

e) $(-6123), (-6321), (-6213), (-6231), (-6312), (-6132)$
 $(-6321), (-6312), (-6231), (-6213), (-6132), (-6123)$

2) Put the following integers into the grid.

(-7) (-10) (-15) (-6) (-20) (-3) (-18) (-10) (-8) (-4)
 (-2) (-5) (-8) (-1) (-9) (-12) (-9) (-5) (-2) (-5)

	Least					Greatest
Least 		(-20)	(-15)	(-9)	(-6)	(-5)
		(-18)	(-10)	(-8)	(-5)	(-4)
		(-12)	(-9)	(-7)	(-3)	(-2)
	Greatest	(-10)	(-8)	(-5)	(-2)	(-1)

